register themselves and their services with a centralized directory (for example, Microsoft Active Directory), which can be queried by other network devices seeking such services.

Recall that even though the application layer is numbered as Layer 7, it is at the top of the OSI stack because its networking functions are closest to the end user.

The TCP/IP Stack

The ISO developed the OSI reference model to be generic, in terms of what protocols and technologies could be categorized by the model. However, most of the traffic on the Internet (and traffic on corporate networks) is based on the TCP/IP protocol suite. Therefore, a more relevant model for many network designers and administrators to reference is a model developed by the U.S. Department of Defense (DoD). This model is known as the *DoD model*, or the ***TCP/IP stack***.

NOTE An older protocol known as Network Control Protocol (NCP) was similar to TCP/IP. NCP was used on ARPANET (the predecessor to the Internet), and it provided features like those offered by the TCP/IP suite of protocols on the Internet, although they were not as robust.

Layers of the TCP/IP Stack

The TCP/IP stack has only four defined layers, as opposed to the seven layers of the OSI model. Figure 1-15 contrasts these two models.

Key Topic

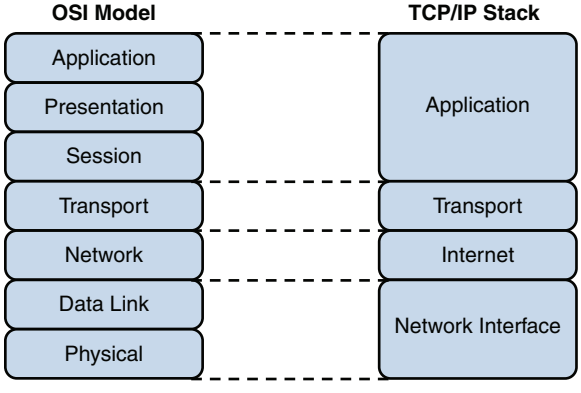


Figure 1-15 TCP/IP Stack

The TCP/IP stack is composed of the following layers:

- **Network interface:** The TCP/IP stack's ***network interface layer*** encompasses the technologies offered by Layers 1 and 2 (the physical and data link layers) of the OSI model.

NOTE Some literature refers to the network interface layer as the *network access layer*.

- **Internet:** The ***Internet layer*** of the TCP/IP stack maps to Layer 3 (the network layer) of the OSI model. Although multiple routed protocols live at the OSI model's network layer, the Internet layer of the TCP/IP stack focuses on IP as the protocol to be routed through a network. Figure 1-16 shows the format of an IP version 4 (IPv4) packet.

Key Topic

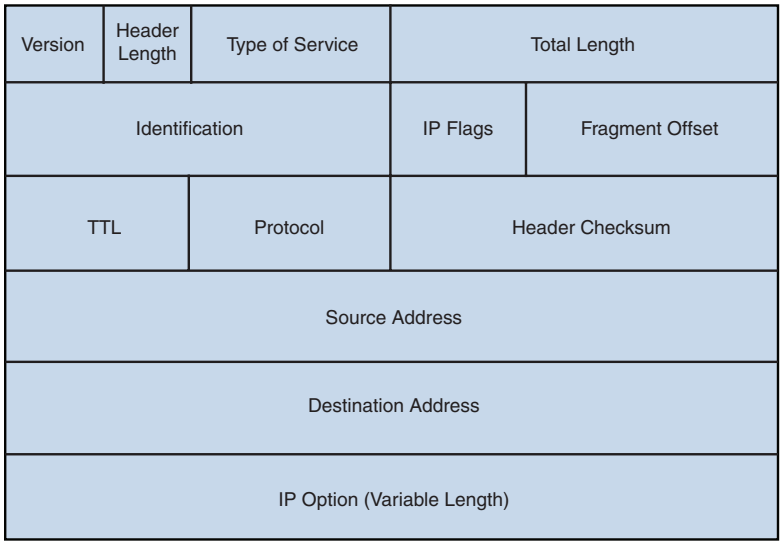


Figure 1-16 IPv4 Packet Format

Notice that there are fields in the IP packet header for both source and destination IP addresses. The Protocol field shows the transport layer protocol from which the packet was sent or to which the packet should be sent. Also of note is the Time-to-Live (TTL) field. The value in this field is decremented by 1 every time this packet is routed from one IP network to another (that is, when it passes through a router). If the TTL value ever reaches 0, the packet is discarded from the network. This behavior helps prevent routing loops, and we will revisit this concept in Chapter 2, "Networking Appliances, Applications,

and Functions." As a common practice, the OSI layer numbers 1, 2, and 3 are still used when referring to physical, data link, and network layers of the TCP/IP stack, even though the TCP/IP stack does not explicitly separate the physical and data link layers.

- **Transport:** The ***transport layer*** of the TCP/IP stack maps to Layer 4 (the transport layer) of the OSI model. The two primary protocols found at the TCP/IP stack's transport layer are TCP and UDP.

  Figure 1-17 details the structure of a TCP segment. Notice the fields for source and destination ports. As described later in this chapter, these ports identify to which upper-layer protocol data should be forwarded or from which upper-layer protocol the data is being sent.

Key Topic

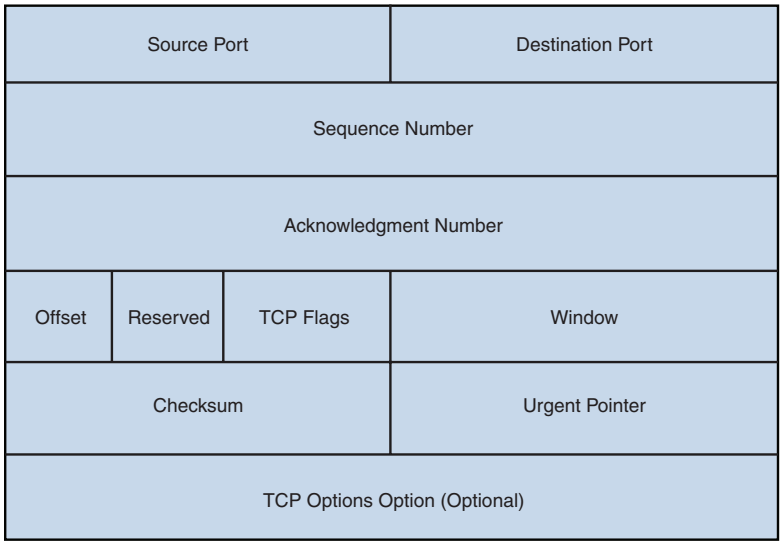


Figure 1-17 TCP Segment Format

Also notice the field for window size. The value in this field determines how many bytes a device can receive before expecting an acknowledgment. As previously described, this feature offers flow control.

The header of a TCP segment also contains sequence numbers for segments. With sequence numbering, if segments arrive out of order, the recipient can put them back in the proper order based on the sequence numbers.

The acknowledgment number in the header shows the next sequence number the receiver expects to receive. This is a way for the receiver to let the sender know that all segments up to and including that point have been received. Due

to the sequencing and acknowledgments, TCP is considered to be a *connection-oriented* transport layer protocol.

NOTE You might have noticed that both the ***IP header*** and the ***TCP header*** make use of a Flags field. Including this field is a very common technique in networking to permit the data unit to convey specific settings. For example, the IP header uses the IP Flags field (3 bits) to help manage (or prevent) fragmentation. ***TCP flags*** are used to indicate a particular connection state or provide additional information. They are often used for troubleshooting purposes or to control how a particular connection is handled.

Figure 1-18 presents the structure of a UDP segment. UDP is a connectionless, unreliable protocol. UDP lacks the sequence numbering, window size, and acknowledgment numbering present in the header of a TCP segment. The UDP segment's header simply contains source and destination port numbers, a UDP checksum (which is an optional field used to detect transmission errors), and the segment length (measured in bytes).

Key Topic

Source Port	Destination Port
UDP Length	UDP Checksum

Figure 1-18 UDP Segment Format

Because a ***UDP header*** is much smaller than a TCP header, UDP is a good candidate for the transport layer protocol for applications that need to maximize bandwidth and do not require acknowledgments (for example, audio or video streams).

- **Application:** The biggest difference between the TCP/IP stack and the OSI model is at the TCP/IP stack's ***application layer***. This layer addresses concepts described by Layers 5, 6, and 7 (the session, presentation, and application layers) of the OSI model.

With the reduced complexity of a four-layer model like the TCP/IP stack, network designers and administrators can more easily categorize a given networking technology into a specific layer. For example, although SIP was shown earlier as a session layer protocol within the OSI model, you would have to know more about the behavior of SIP to properly categorize it in that model. However, with the TCP/IP stack, you could quickly figure out that SIP is a higher-level protocol that gets

encapsulated inside TCP, and you could thus classify SIP in the application layer of the TCP/IP stack.

Common Application Protocols in the TCP/IP Stack

Application layer protocols in the TCP/IP stack are identifiable by unique port numbers. For example, when you enter a web address in an Internet browser, you are (by default) communicating with that remote web address using TCP port 80. Specifically, Hypertext Transfer Protocol (HTTP), which is the protocol used by web servers, uses TCP port 80. Therefore, the data you send to that remote web server has a destination port number of 80. That data is encapsulated into a TCP segment at the transport layer. That segment is further encapsulated into a packet at the Internet layer and sent out on the network using an underlying network interface layer technology such as Ethernet.

NOTE Thanks to awareness of network security today, you do not see HTTP (port 80) actually being used on the Internet very much anymore. It has been replaced with a secured version, HTTPS, which uses TCP port 443 in its operation.

Consider the example illustrated in Figure 1-19. When you send traffic to the remote website, the packet you send out to the network needs not only the destination IP address (172.16.1.2 in this example) of the web server and the destination port number for HTTP (that is, 80) but also the source IP address of your computer (10.1.1.1 in this example). Because your computer is not acting as a web server, its port is not 80. Instead, your computer selects a source port number greater than 1023. In this example, let's imagine that the client PC selects the source port 1248.

Key Topic

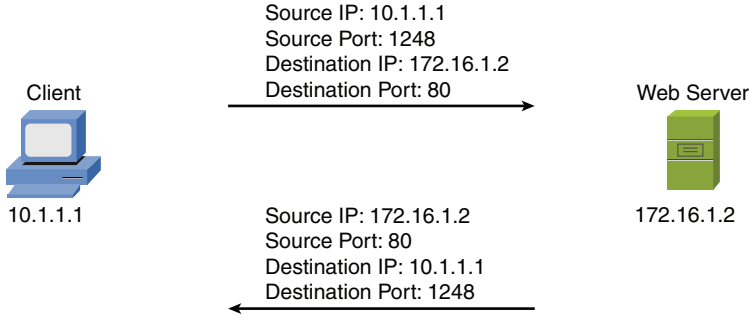


Figure 1-19 Example: Port Numbers and IP Addresses

Notice that when the web server sends content back, the IP addresses and port numbers have now switched, with the web server as the source and your PC as the

destination. With both source and destination port numbers, along with source and destination IP addresses, two-way communication becomes possible.

NOTE Ports numbered 1023 and below are called *well-known* ports, and ports numbered above 1023 are called *ephemeral* ports. The maximum value of a port is 65535. Well-known port number assignments are available at https://www.iana.org/assignments/port-numbers.

Real-World Case Study

Bob, a manager of the networking team at Acme, Inc., is paying extra attention to the specific words he uses as he talks to his team in preparation for the implementation of the network. When referring to transport protocols such as the connection-oriented TCP and the connectionless UDP, the word Bob uses to describe those protocol data units is *segment*.

In discussing the applications that Acme, Inc. will be using over its network, Bob notes that many of these applications will be using TCP at the transport layer. This includes HTTPS for secure web traffic and Simple Mail Transfer Protocol (SMTP) and Internet Message Access Protocol (IMAP) for email services.

The company will use the Secure Shell (SSH) protocol, which also uses TCP at the transport layer, as a secure method to remotely connect to and manage its network devices. A common connectionless UDP protocol is Domain Name System (DNS), which will be used thousands of times a day to translate a friendly name like https://www.pearson.com to an IP address that is reachable over the network. Another protocol based on UDP that will be used often is Dynamic Control Host Protocol (DHCP), which assigns client computers on the network an IP address that is required for sending and receiving Layer 3 packets.

For the traffic on the LAN, the Ethernet cables and electronic signals being sent as bits going over those cables represent Layer 1 from an OSI perspective. On the LAN, they will be using Ethernet technology, and as a result, the Layer 2 frames that are sent on the LAN will be encapsulated and sent as Ethernet Layer 2 frames.

For datagrams being sent across the serial WAN connections provided by the service provider, it is likely that protocols and technologies such as High-Level Data Link Control (HDLC), Layer 2 Tunneling Protocol (L2TP), Point-to-Point Protocol (PPP), Point-to-Point Tunneling Protocol (PPTP), Spanning Tree Protocol (STP), and virtual LANs (VLANs) are in operation at Layer 2 (the data link layer). On both the LAN and the WAN, at Layer 3 (the network layer), IPv4 will be used